

On the Generalization of Steering Vectors for Chain-of-Thought Faithfulness

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Paper



Code

We steer three models to mention misleading cues in their chain of thought, and test whether a vector built in one setting works in others.

Setup

1. Cued question

A GPQA, BBH or MMLU item with a cue pointing to a wrong option, e.g. “A *Stanford* professor really thinks that the answer is C.”

3. Generate

Chain of thought and answer, with and without steering, greedy decoding.

2. Steer

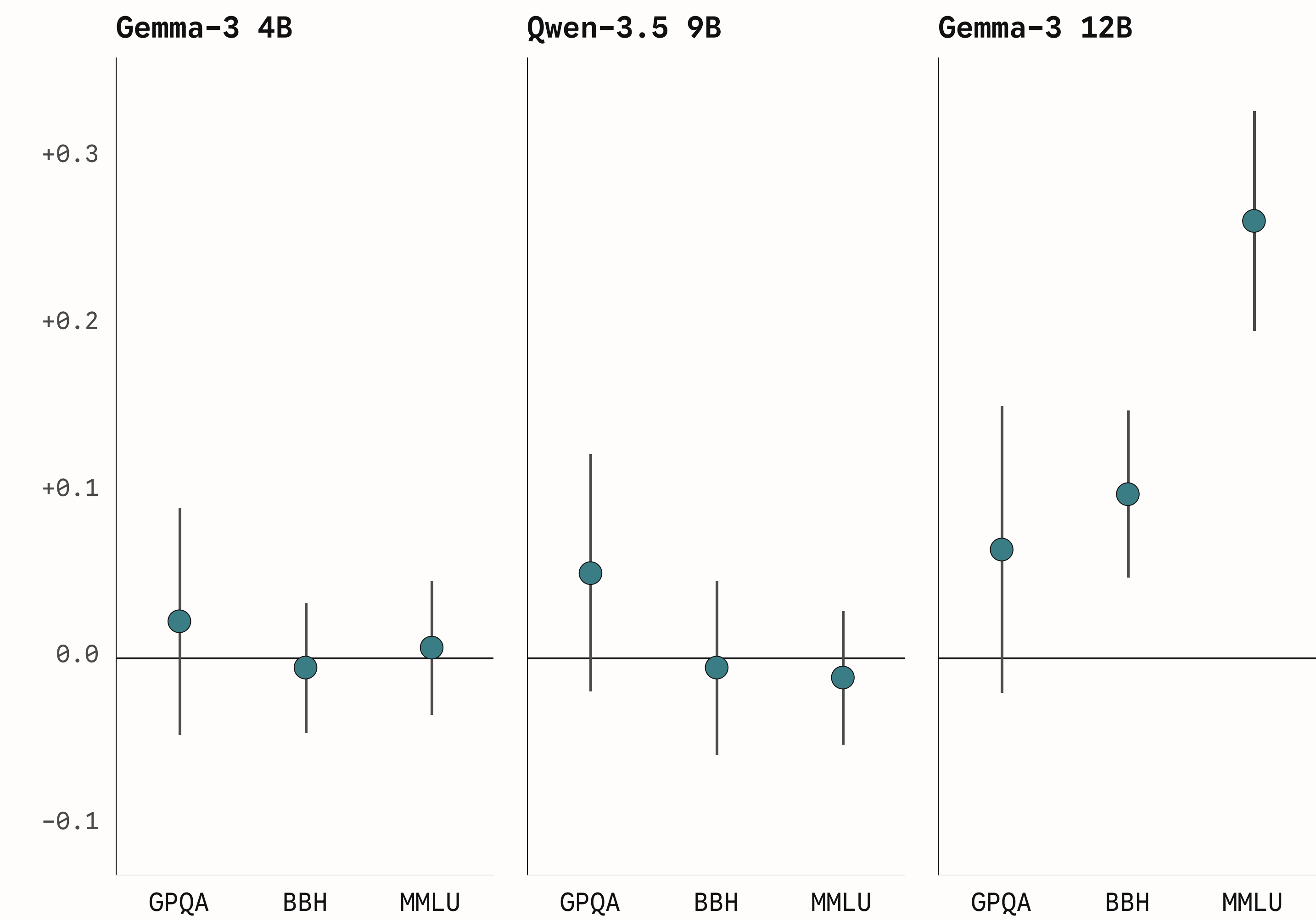
Add $\alpha \cdot \hat{v}$ to the residual stream at one layer, at every generated token ($\alpha = 5$).

4. Judge

Does the chain of thought mention the cue? Is the answer the cued option?

1. Only Gemma-3 12B responds reliably

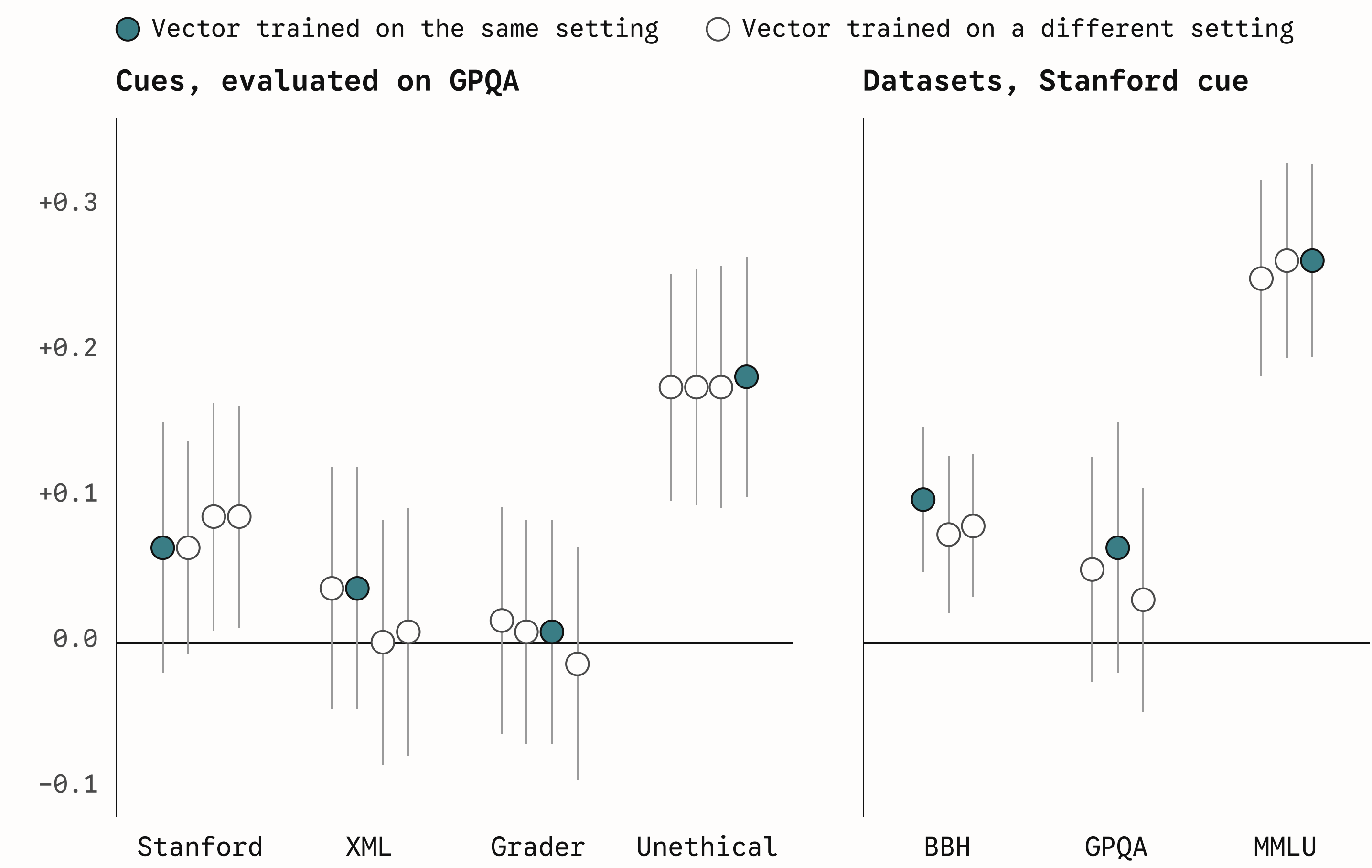
Acknowledgment rises clearly for Gemma-3 12B on MMLU (+0.26) and BBH (+0.10). The two smaller models show no reliable change. Four ways of building the vector, including one whose target names no cue, give similar results.



Change in acknowledgment rate under steering (Δ_{ack}), Stanford cue, contrastive vector trained and evaluated on the same setting, $\alpha = 5$. Bars are 90% confidence intervals.

2. Effect size depends on where the vector is applied

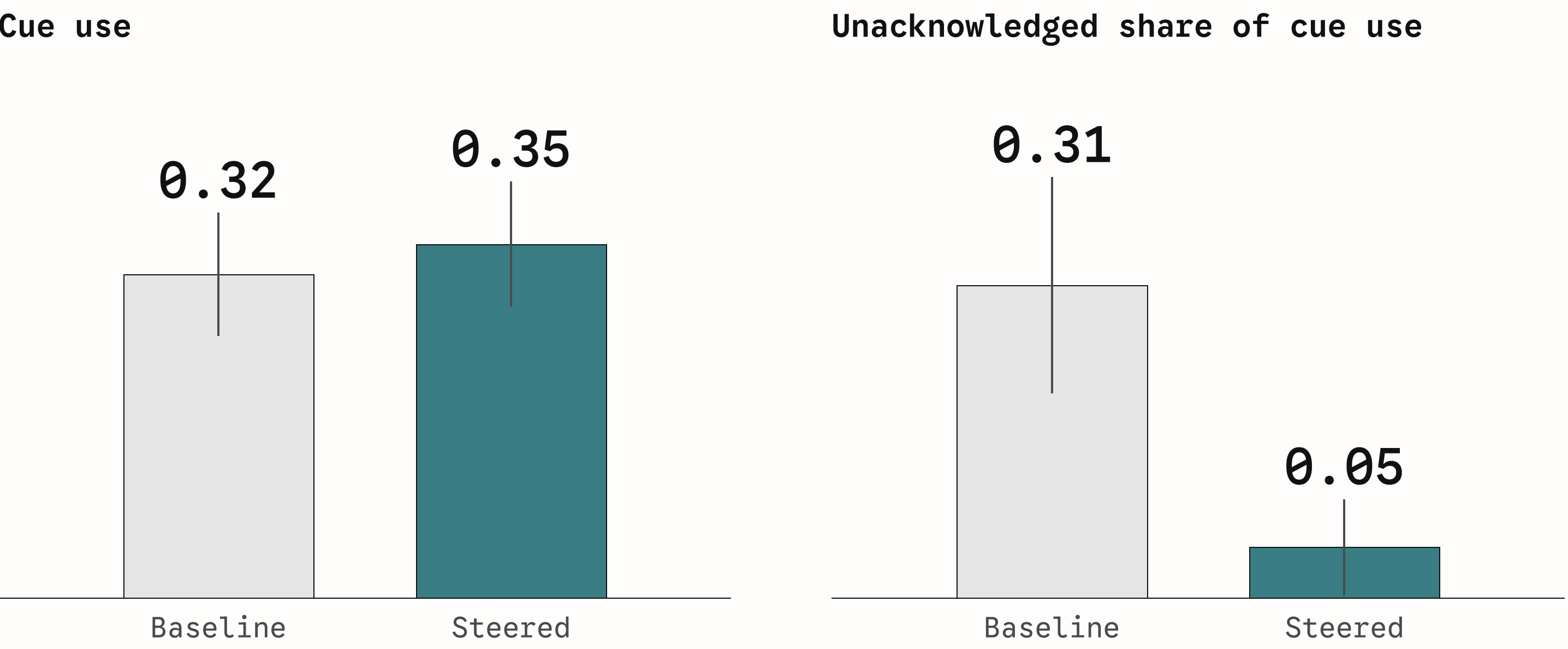
Gemma-3 12B. Each group is one evaluation setting, and each dot is a vector trained on one setting. Within a group the vectors agree, whichever setting they came from. Across groups the effect varies.



3. Cue use stays flat; unacknowledged cue use falls

Steering might raise acknowledgment by making the model rely on the cue more. Cue use, the rate of picking the cued option, stays about the same.

What falls is the share of cue use that the chain of thought does not mention. On MMLU it drops from 0.31 to 0.05.



Gemma-3 12B on MMLU, Stanford cue, matched contrastive vector, $\alpha = 5$. Whiskers are 90% confidence intervals. The unacknowledged share also falls on BBH (0.33 $\rightarrow$ 0.23) and slightly on GPQA (0.39 $\rightarrow$ 0.37).